INTRO TO THE CLIMATE SYSTEM

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CEVE 543

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HAVE YOU TAKEN A CLASS ON CLIMATE?

Set your Poll Everywhere screen name to your Rice NetID so your answers are recorded.

PollEV.com/jdossgollin

Context

Last Week: Independent and identically distributed (IID)

This Week: Why and how the climate system violates these assumptions

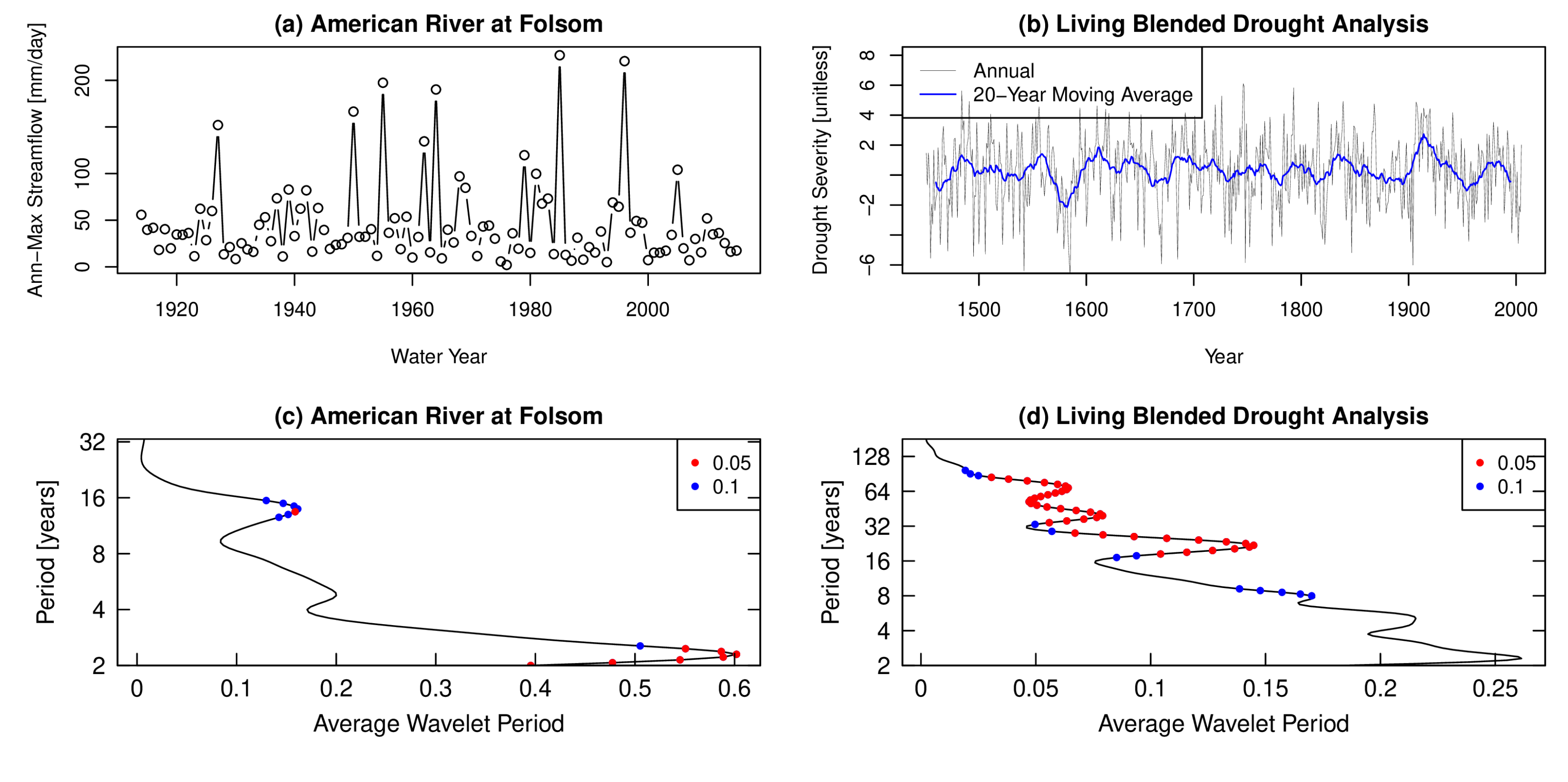
Exploratory Data Analysis

The climate system

What weather patterns generate extremes

El Niño-Southern Oscillation (ENSO) as a case study

MULTI-SCALE VARIABILITY



American River at Folsom, California (Doss-Gollin et al., 2019)

WHAT DO YOU SEE?

Set your Poll Everywhere screen name to your Rice NetID so your answers are recorded.

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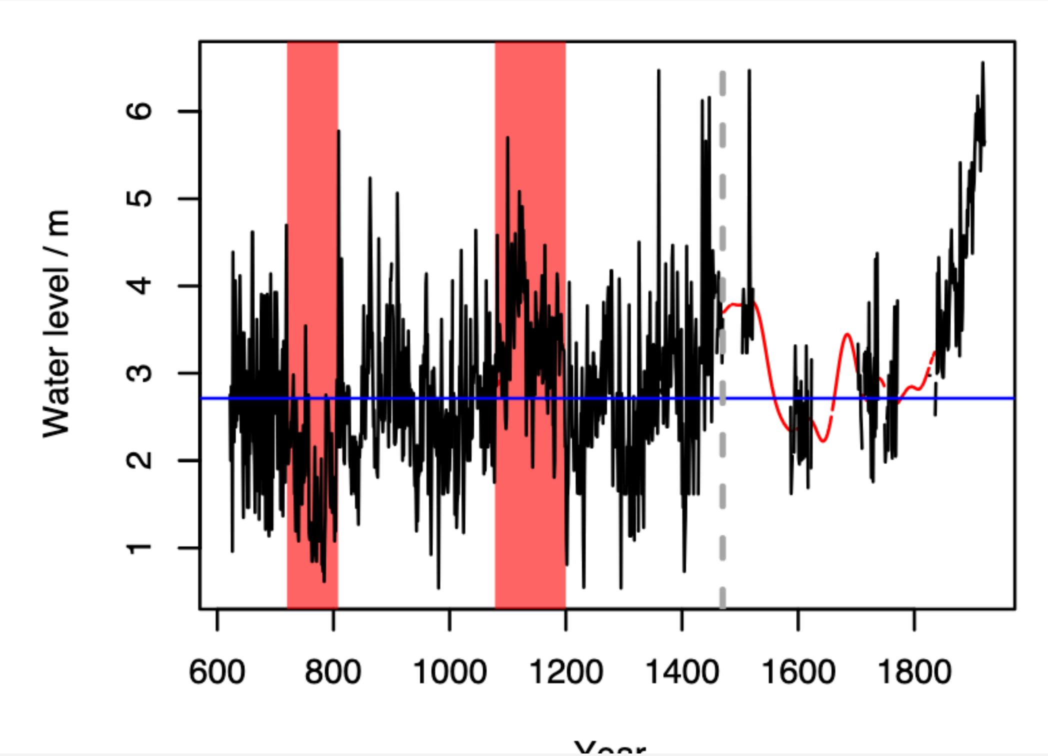
1300 YEARS OF NILE ANNUAL MINIMA



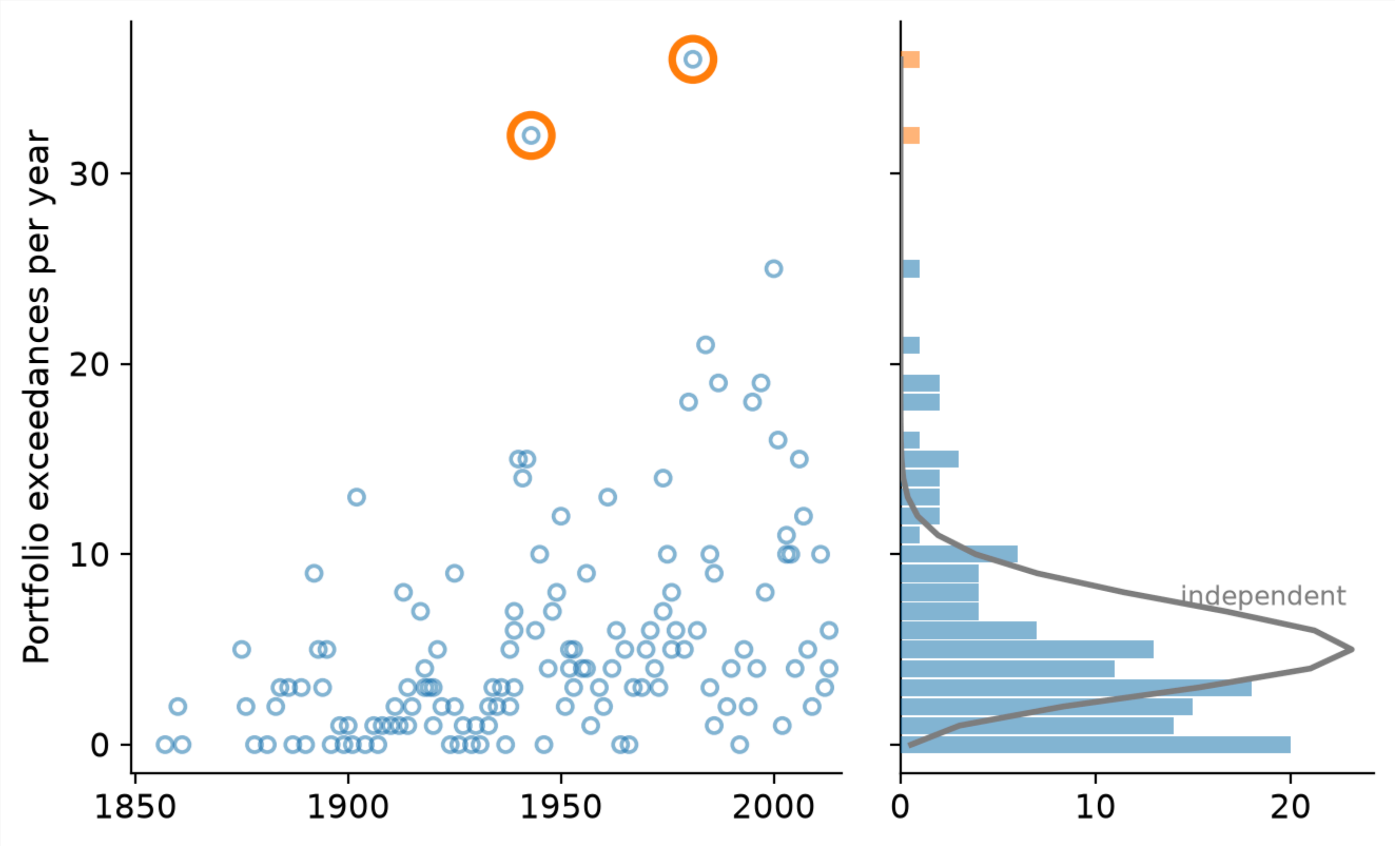
Nile annual minima, 622-1921 AD (Ghil et al., 2011, Fig. 1)

CLIMATE RISKS CLUSTER IN TIME

Note

Figure shown in class: North American megadroughts over the past 1200 years (Cook et al., 2010)

CLIMATE RISKS CLUSTER IN SPACE



Yearly count of 30-day extreme rainfall exceedances across 40 mining assets, compared to the count expected under independence (Bonnafous et al., 2017)

HOW DO WE DESCRIBE THIS?

- Clustering
- Persistence
- Memory
- Regimes

Not IID

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RESERVOIRS, FLUXES, AND FEEDBACKS

Note

Figure shown in class: The climate system (IPCC TAR, Ch. 1, Fig. 1.1)

MODELLING THE CLIMATE SYSTEM

Note

Figure shown in class: Coupled climate model schematic (NOAA GFDL)

A FLUID ON A ROTATING SPHERE

Note

Figure shown in class: Equations of motion for a fluid on a rotating sphere (Vallis, 2006)

WEATHER VERSUS CLIMATE

Weather

- we can make (and verify!) *forecasts*
- Initial value problem

Climate

- “the statistics of weather”
- we can make, but not verify, *projections*
- Boundary value problem

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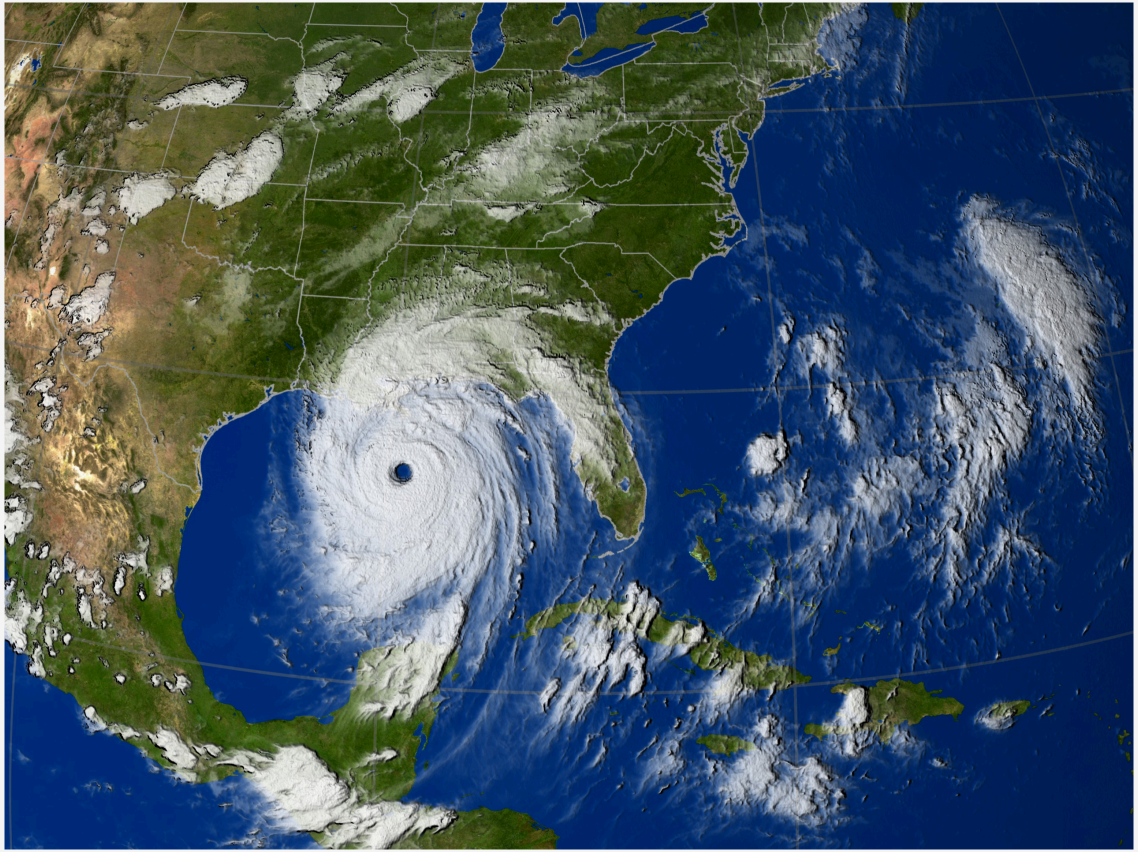
Atmospheric Rivers

i Note

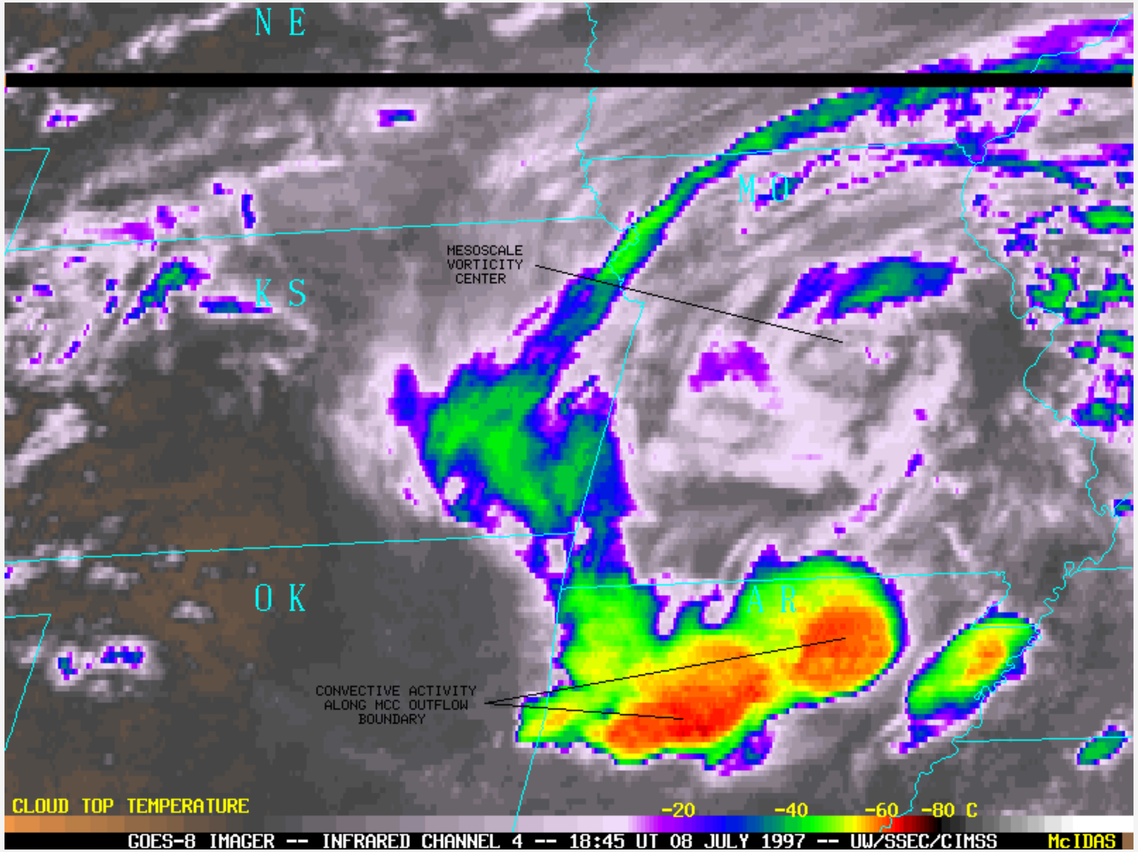
Figure shown in class: Atmospheric river moisture transport (Gimeno et al., 2014)

TROPICAL CYCLONES



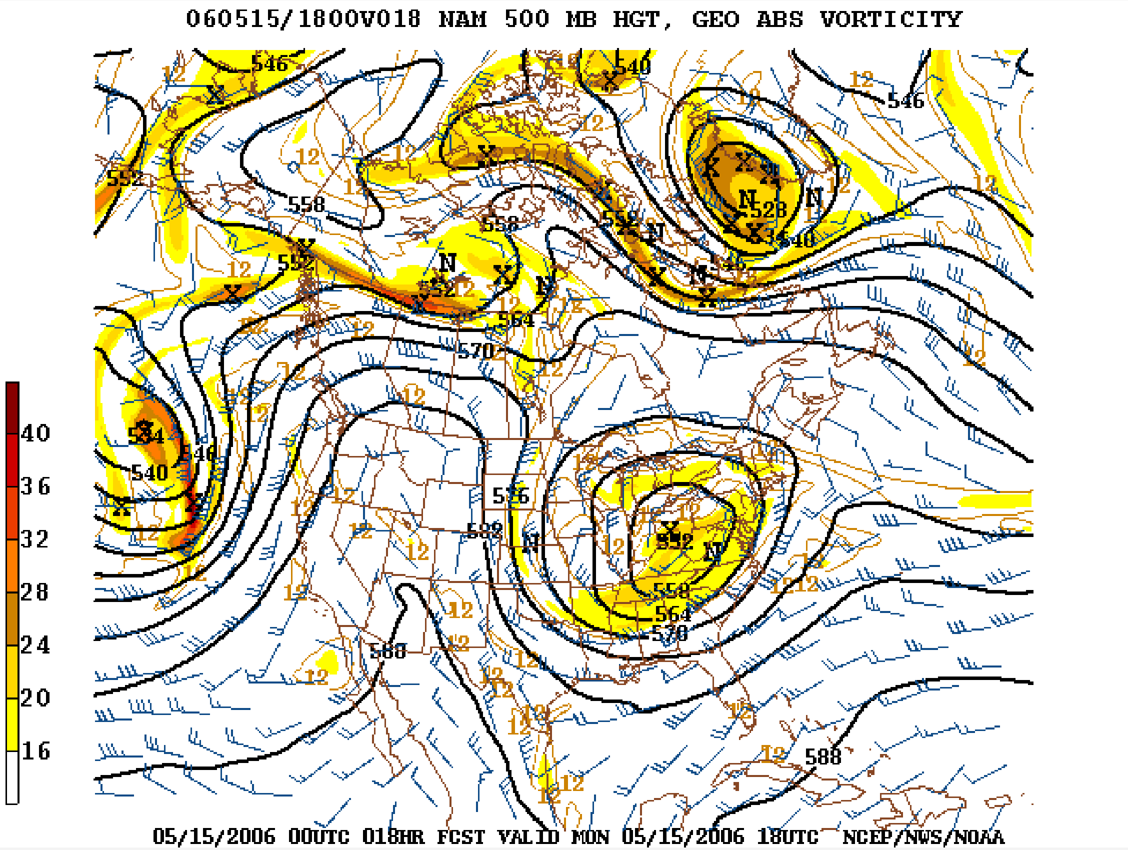
Hurricane Katrina, 2005 (NASA/NOAA)

MESOSCALE CONVECTIVE SYSTEMS



Mesoscale convective complex, Great Plains, 1997 (CIMSS/NOAA)

BLOCKING



Omega block over North America (NCEP/NWS/NOAA)

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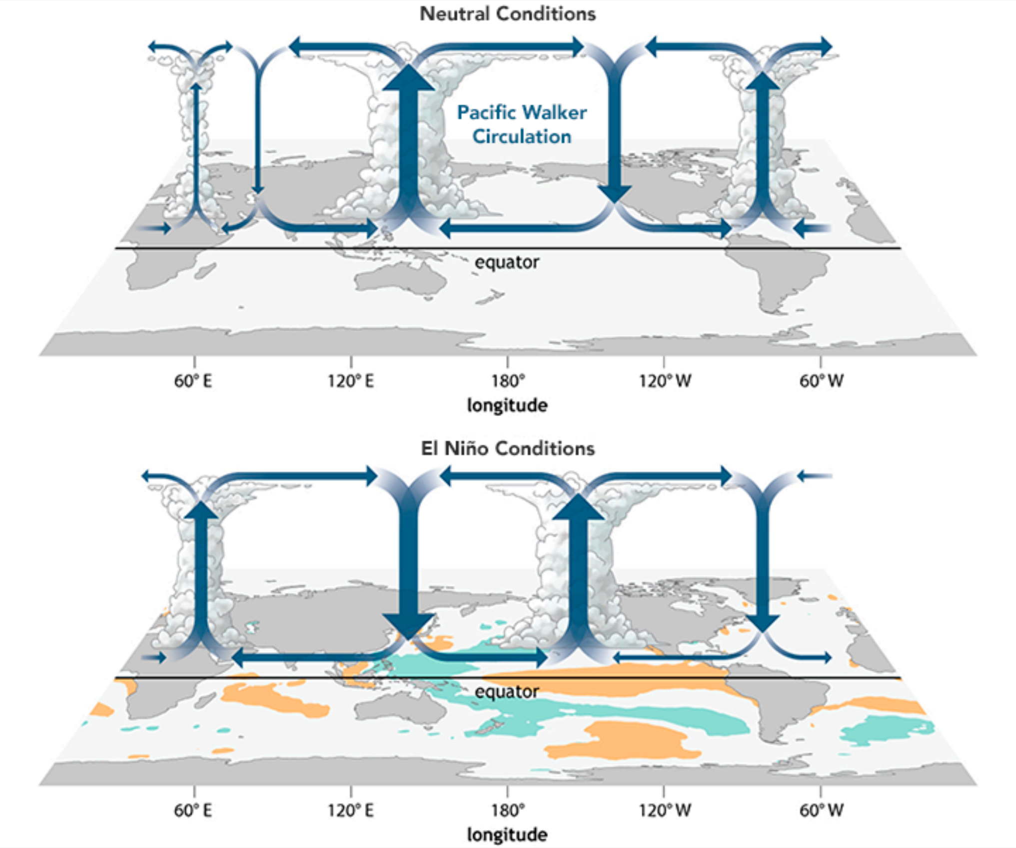
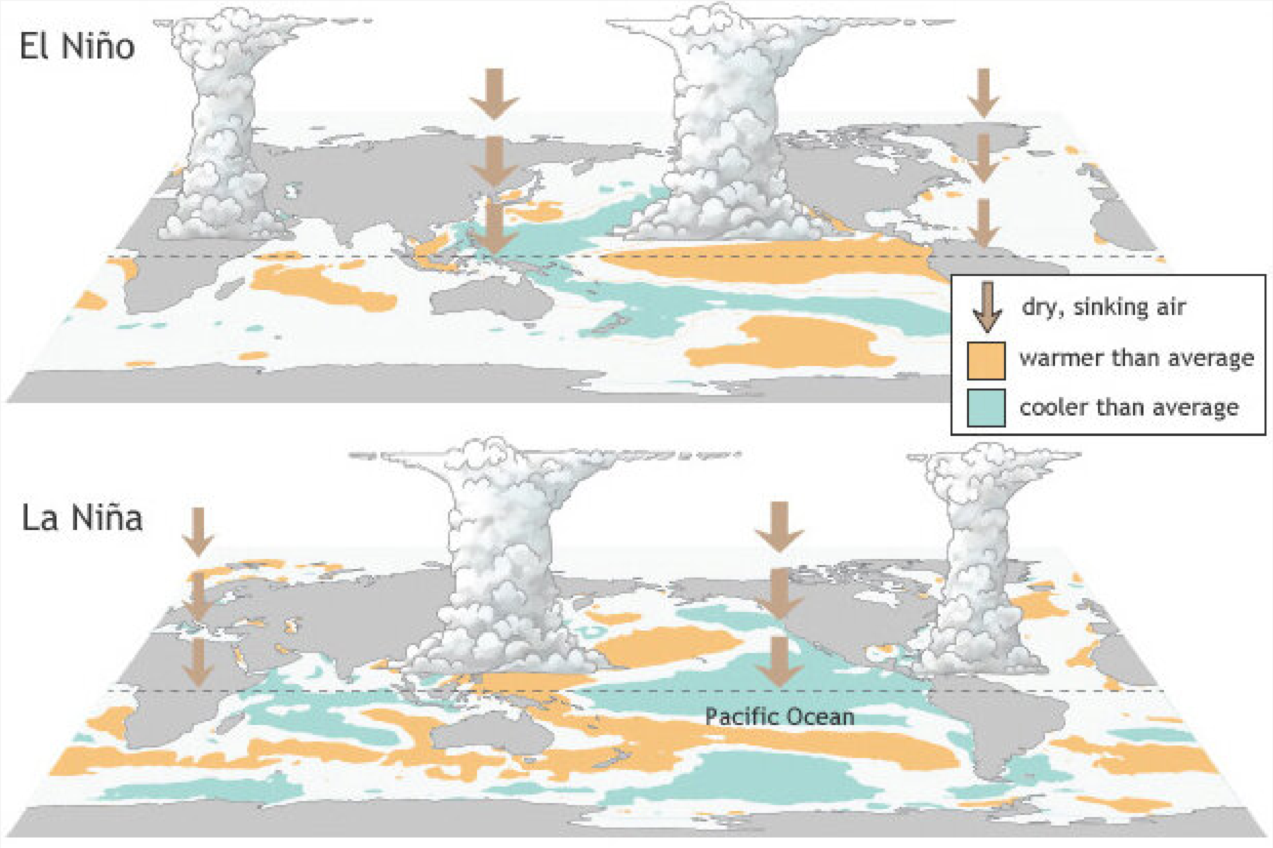
THE WALKER CIRCULATION



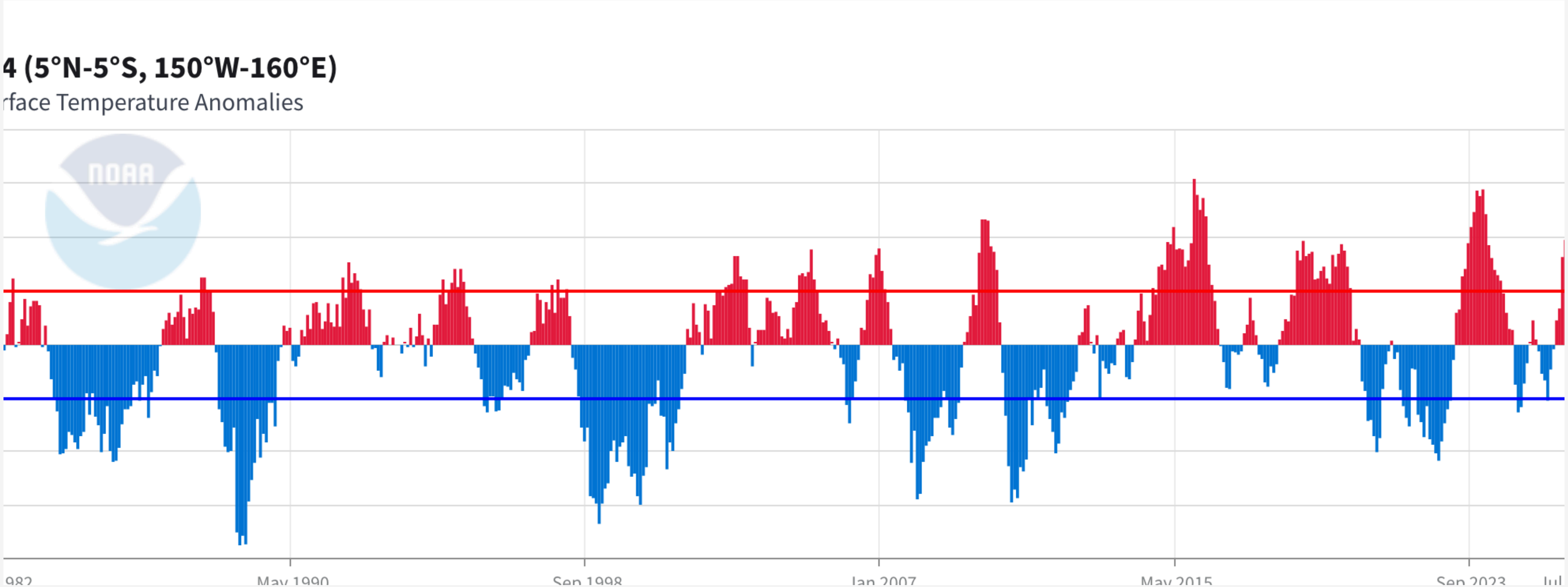
Normal and El Niño conditions in the tropical Pacific (NOAA Climate.gov)

EL NIÑO VERSUS LA NIÑA



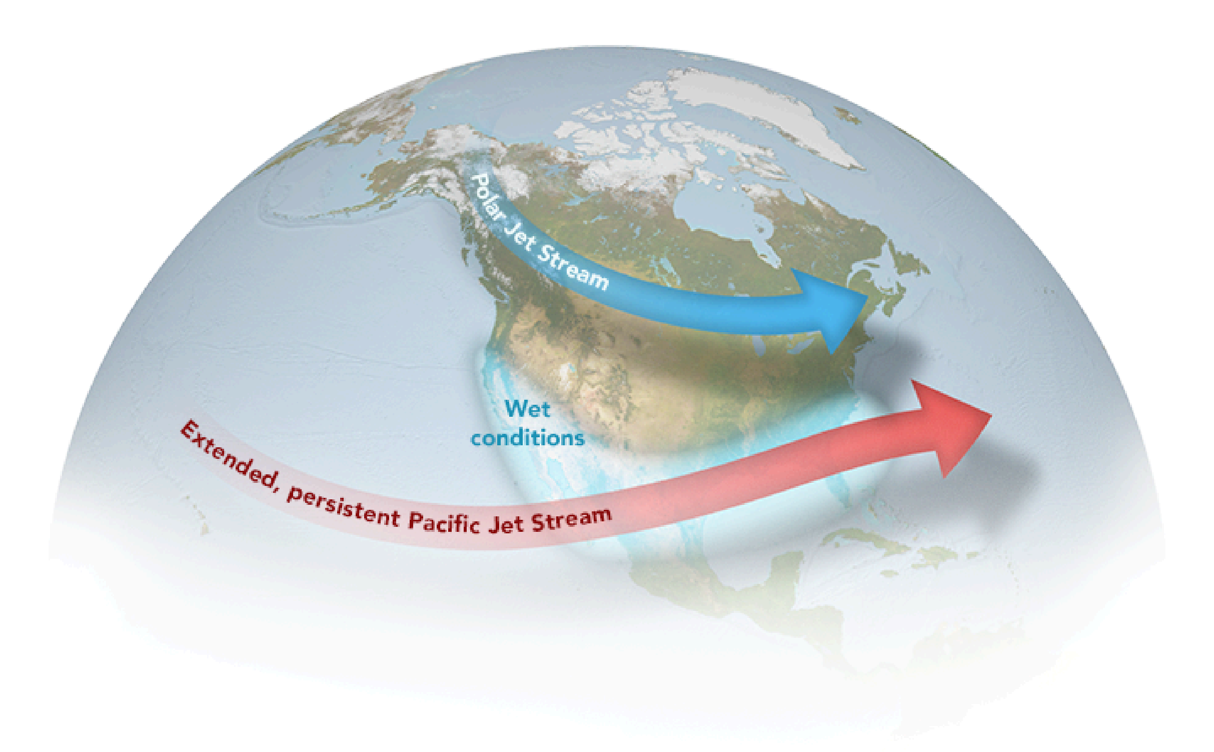
El Niño and La Niña SST and convection patterns (NOAA Climate.gov)

RIGHT NOW



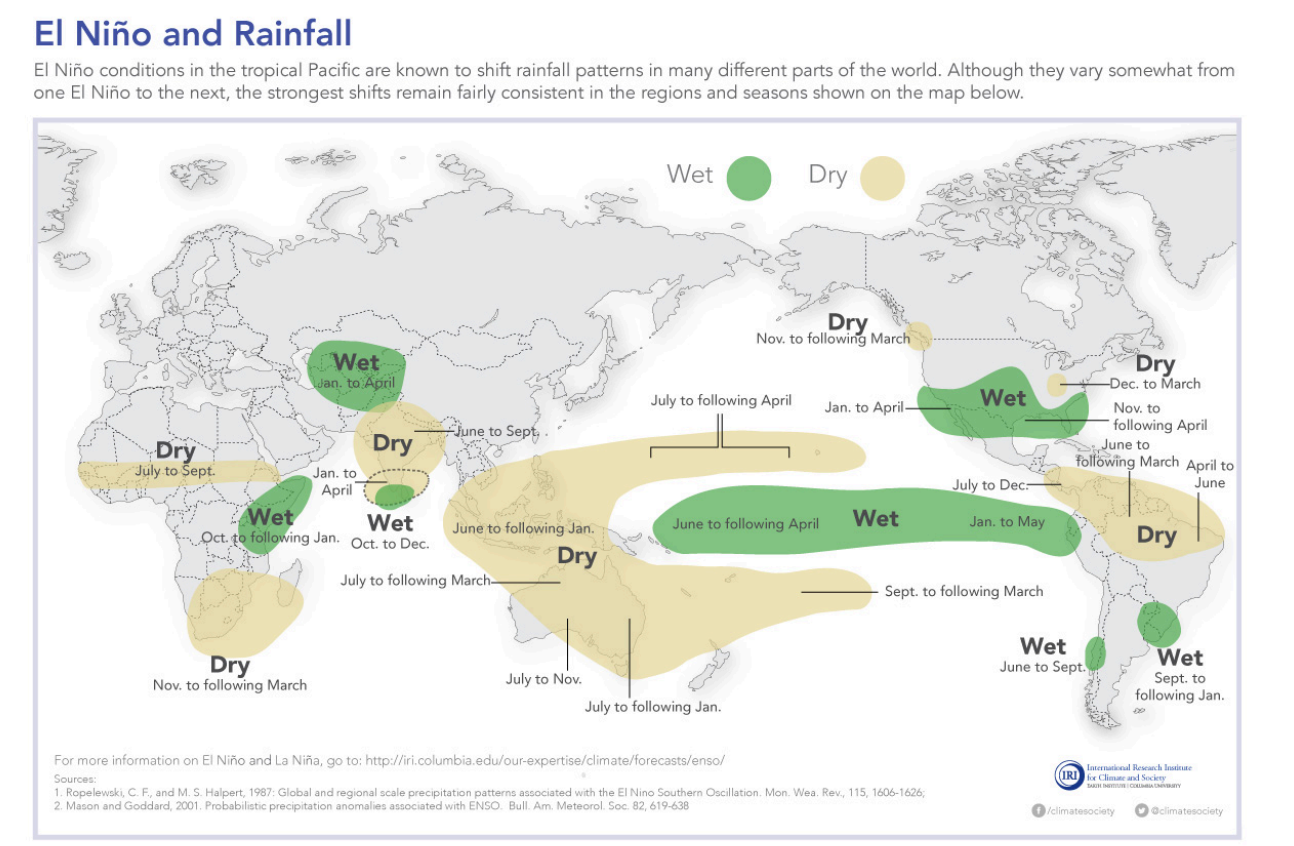
Niño 4 SST anomalies, 1982-2026 (NOAA CPC)

IMPACTS ON THE JET STREAM



El Niño extends the Pacific jet stream eastward (NASA Earth Observatory, Joshua Stevens)

EL NIÑO AND RAINFALL

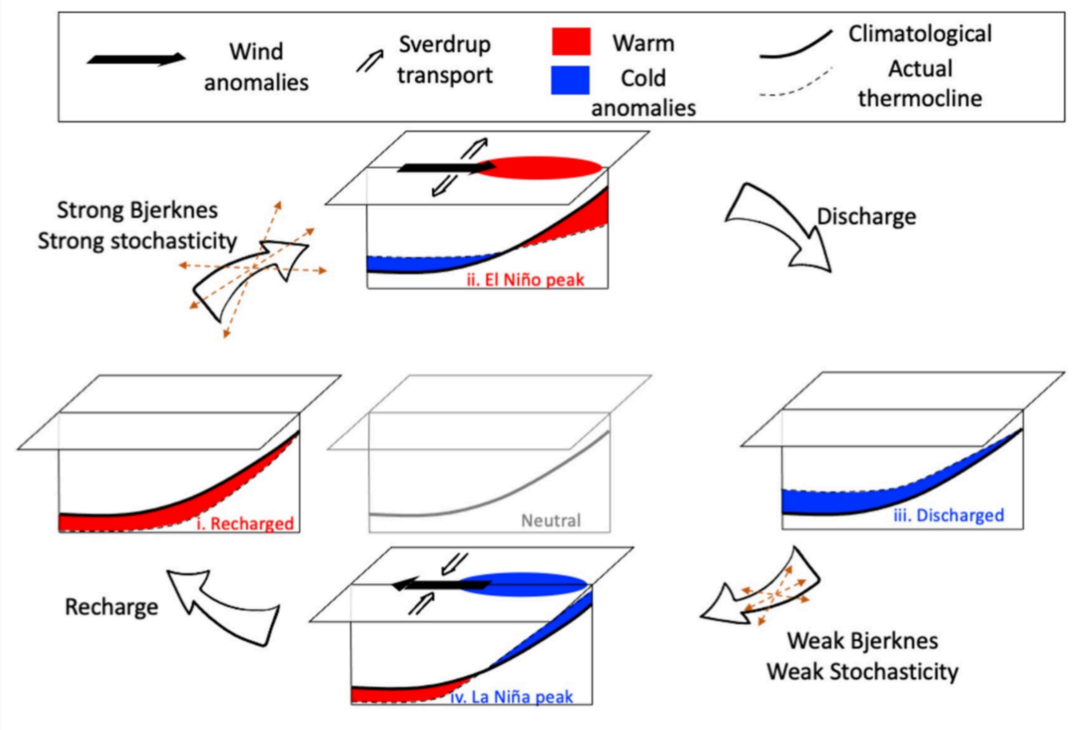


El Niño shifts rainfall patterns globally; the strongest shifts are consistent from event to event (IRI, Columbia University)

THE MECHANISM

The Bjerknes feedback: warm sea surface temperatures (SSTs) weaken the trade winds, which deepens the thermocline (the warm-cold boundary) in the east, which warms SSTs further (Vialard et al., 2025, p. 4).

The recharge oscillator captures this as a four-phase cycle.



Recharge oscillator (Vialard et al., 2025, Fig. 7)

OTHER MODES OF VARIABILITY

Mode	Timescale	Region
Madden-Julian Oscillation	30-60 days	Tropics, modulates ENSO onset
ENSO	2-7 years	Tropical Pacific, global reach
North Atlantic Oscillation	Weeks to decades	North Atlantic, European weather
Pacific Decadal Oscillation	20-30 years	North Pacific, modulates ENSO
Atlantic Multidecadal Oscillation	60-80 years	North Atlantic, hurricane activity

DISCUSSION

- What does multi-scale variability mean for hazard analysis based on a 50-year record?
- Can we use observations to relate ENSO phase to local extremes?